



Project Title: Swift signals

Guide Name: Prof. AVADESH KUMAR

Course code: ECS2002

Team Members:

- 1.TELLAKULA DATHRIJA (23BCE8917)
- 2.BAINABOINA SREE SOWMYA (23BCE9879)
3. JINKA GAYATHRI (23BCE9581)
4. KELAVATHU RUKMINI BAI (23BCE20189)
5. VARIKUTI PALLAVI (23BCE20076)
6. KANIKE KUNDANA (23BCE9702)

Abstract:

In Today's rapidly urbanizing world, traffic congestion and delays have become significant challenges, impacting both daily commuters and emergency response teams. This project presents a smart traffic management system designed to optimize traffic flow and prioritize emergency vehicles. The system uses a network of IR sensors to monitor vehicle density across four directions at an intersection, enabling adaptive control of traffic lights to reduce congestion based on real-time data. Additionally, an RF module detects the presence of emergency vehicles, allowing the system to give priority by immediately switching the light to green in the relevant direction. This approach aims to enhance traffic efficiency, minimize wait times, and facilitate faster emergency responses, thereby contributing to safer and more efficient urban transportation.

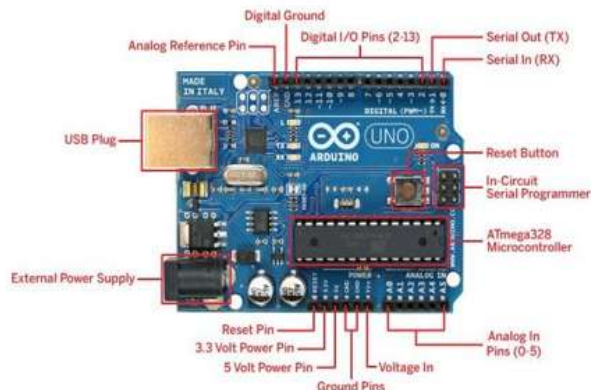
INDEX

Contents	page no
1. List of figures	3
2. List of tables	4
3. Introduction	5
4. Background	6
5. Problem Definition	6
6. Objectives	7
7. Methodology	7
8. Results and Discussion	8
9. Conclusion and Future scope	9
10. References	9
11. Codes in appendix	10

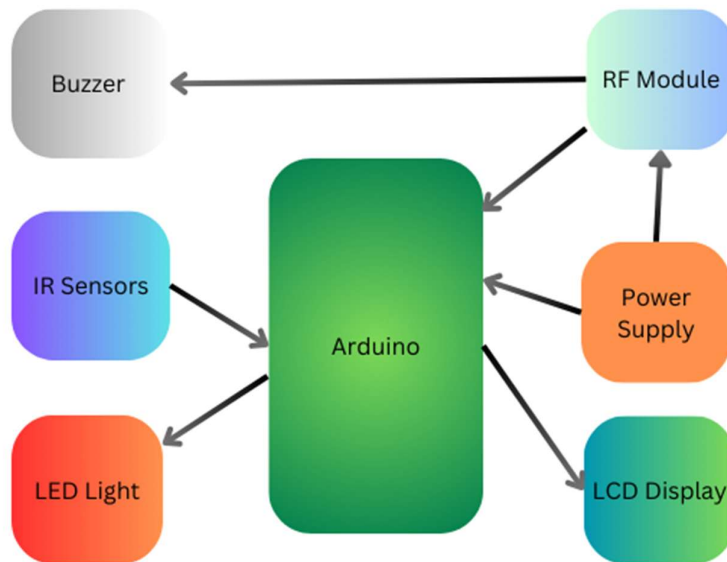
1.LIST OF FIGURES:

FIGURE No.	TITLE	PAGE NO.
1.1	Architecture Diagram	3
1.2	Block Diagram	4

1.1 Architecture Diagram



1.2 Block Diagram



2.List of Tables: Arduino uno Specifications

Microcontroller	AT mega 328
Operating voltage	5V
Input voltage (recommend)	5V
Input voltage(limit)	3V-8V
Digital I/O pins	2-13
Analog input pins	0-5
DC current per I/O pin	20mA
DC current for 3.3V pin	50mA
SRAM	2KB(AT mega 328P)
EEPROM	1kB(ATmega328P)

3.Introduction:

With increasing urbanization and the rise in vehicle numbers, traffic congestion has become a pressing issue for cities worldwide. Conventional traffic light systems often rely on fixed time cycles that do not adapt to real-time traffic conditions, leading to unnecessary delays and prolonged congestion. Moreover, these traditional systems lack a mechanism for prioritizing emergency vehicles, which can result in delays for ambulances, fire trucks, and other critical response units.

To address these challenges, this project proposes that utilizes IR sensors and RF mod Swift signals rules. The IR sensors are deployed in four directions at a busy intersection to detect the density of vehicles in each lane. By analyzing the data from these sensors, the system dynamically adjusts the duration of green and red lights, optimizing traffic flow based on actual conditions rather than preset cycles. Furthermore, an RF module is integrated into the system to detect emergency vehicles, which emit a specific signal when approaching the intersection. Upon detection, the system instantly prioritizes the direction of the emergency vehicle, granting it a green light for swift passage and ensuring that it encounters minimal delay.

Swift signals solution aims to provide a more efficient, adaptive, and responsive approach to urban traffic control. By reducing congestion and allowing for emergency vehicle prioritization, the system enhances both road safety and traffic flow, making it a vital component in the infrastructure of modern smart cities.

4. Background:

Increasing urban vehicle numbers and limited road infrastructure, traffic congestion has become a significant urban issue worldwide. Traditional traffic light systems operate on fixed cycles, often disregarding real-time traffic conditions, which leads to delays, higher fuel consumption, and emissions.

The limitations of these systems become critical for emergency vehicles, like ambulances and fire trucks, which face delays at intersections due to lack of prioritization. This can result in slower emergency response times, impacting life-saving efforts.

Technological advancements now enable smarter, adaptive traffic solutions. Systems using sensors and wireless communication can monitor real-time traffic, prioritize emergency vehicles, and dynamically control lights. By integrating IR sensors to detect vehicle presence and RF modules to identify emergency vehicles, a smart traffic management system can reduce congestion and improve urban traffic flow, aligning with the smart city goal of efficient, responsive infrastructure.

5. Problem Definition:

Traffic congestion at intersections is a major issue in urban areas, leading to delays and increased travel time. Traditional traffic light systems often use fixed timings, which do not account for real-time traffic conditions, resulting in inefficiencies. Additionally, emergency vehicles frequently face delays due to congested intersections.

There is a need for an intelligent traffic management system that adapts to real-time conditions, prioritizes emergency vehicles.

6.Objectives:

- **Optimize Traffic Flow:** Develop a system that uses real-time data from sensors to adjust traffic light timings, reducing congestion at intersections.
- **Enhance Road Safety:** To decrease idle time for vehicles, Leading to lower fuel consumption and reduced greenhouse gas emissions.
- **Prioritize Emergency Vehicles:** Implement a mechanism to detect and prioritize emergency vehicles, providing them with green lights to facilitate quicker response times.

7.Methodology:

There are 4 IR sensors in each direction (East, West, North, South) to detect vehicles density. There is a RF module in the emergency vehicle it gives the signal to RF receiver to detect the emergency vehicle. After connecting all the parts using jumper wires and uploading code in Arduino. The power supply is connected to Arduino.

Firstly 5 seconds is provided to each direction. Each IR sensor detects the vehicles density in each direction and the Arduino Analyse the data and increases time for each direction according to the density and the direction with high density is prioritized. When an emergency vehicle (ambulance, fire truck, police vehicle) approaches, it emits a specific RF signal that is detected by the RF module at the traffic signal.

Upon detecting the RF signal, the controller immediately prioritizes the direction of the emergency vehicle, giving it a green light. The system keeps the other directions on red until the emergency vehicle has cleared the intersection, ensuring a quick passage. After emergency vehicle clearance, system returns to normal traffic management.

8.Result:

1. Improved Traffic Flow: The system effectively adjusted traffic signal durations based on real-time vehicle density, leading to a noticeable reduction in congestion at the intersection. This adaptive approach minimized wait times, especially during peak hours, improving overall traffic efficiency.
2. Reduced Emergency Response Times: The RF module successfully detected approaching emergency vehicles, instantly prioritizing their direction with a green light. This feature enabled emergency vehicles to pass through the intersection quickly, reducing response times in critical situations.
3. Enhanced Safety and Efficiency: The system's ability to manage traffic adaptively improved road safety by reducing instances of heavy congestion and providing clear passage for emergency vehicles, supporting safer and more efficient intersections.

9.Conclusion:

The smart traffic management system utilizes IR sensors and RF modules to represents a significant advancement in urban traffic control. By dynamically adjusting traffic light durations based on real-time vehicle density, it helps to reduce congestion and improve traffic flow. Additionally, the RF module allows the system to prioritize emergency vehicles, ensuring they can navigate intersections quickly and safely, which can be critical in life-saving situations. This solution is a step forward in creating more efficient, responsive, and safer road networks, contributing to a smoother transportation experience in smart cities.

10.References:

1. P. Verma, S. Das, "Smart Traffic Management System: Techniques and Challenges," International Journal of Advanced Research in Computer Science, vol. 9, no. 2, pp. 112-118, 2022.
2. A. Singh, M. Kumar, "Vehicle Detection and Tracking Techniques for Traffic Monitoring," Journal of Traffic and Logistics Engineering, vol. 8, no. 4, pp. 67-75, 2020.
3. S. Patel, R. Bhatt, "Emergency Vehicle Detection and Priority Management in Smart Traffic Systems," Transportation Research Journal, vol. 12, no. 3, pp. 145-151, 2019.
4. D. Smith, "Environmental Impact of Smart Traffic Systems: A Study on Emissions Reduction," Journal of Environmental Technology, vol. 14, no. 1, pp. 50-59, 2020.
5. A. Ahmed, L. Chen, "RF Module Applications in Traffic Control and Emergency Response Systems," International Journal of Embedded Systems, vol. 11, no. 6, pp. 345-352, 2019.

11.Code in Appendix:

```
#include <LiquidCrystal.h>

#include<SoftwareSerial.h>

#include <Adafruit_NeoPixel.h>

SoftwareSerial mySerial(2,3);

#ifdef __AVR__

#include <avr/power.h> // Required for 16 MHz Adafruit Trinket

#endif

#define LED_PIN 7

#define LED_COUNT 4

Adafruit_NeoPixel strip(LED_COUNT, LED_PIN, NEO_GRB + NEO_KHZ800);

int buz =6;

int es=A0;

int ws=A3;

int ns=A2;

int ss=A1;

int rf1=5;

int x=0;

const int rs = 8, en = 9, d4 = 10, d5 = 11, d6 = 12, d7 = 13;

LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

int et,wt,nt,st;

int cnt=0,ecnt=0,wcnt=0,ncnt=0,scnt=0,sts=0,i=0;

void setup()

{

Serial.begin(9600); // Initiate a serial communication

mySerial.begin(9600);

pinMode(rf1,INPUT);
```

```

pinMode(es,INPUT);
pinMode(ws,INPUT);
pinMode(ns,INPUT);
pinMode(ss,INPUT);
pinMode(buz,OUTPUT);
lcd.begin(16,2);
lcd.print(" WELCOME");
strip.begin();    // INITIALIZE NeoPixel strip object (REQUIRED)
strip.setBrightness(100);
    strip.setPixelColor(0, strip.Color(255, 0, 0));
    strip.setPixelColor(1, strip.Color(255, 0, 0));
    strip.setPixelColor(2, strip.Color(255, 0, 0));
    strip.setPixelColor(3, strip.Color(255, 0, 0));
strip.show();      // Turn OFF all pixels ASAP
}
void loop() {
lcd.clear();
cnt=5+ecnt;
et=ecnt;
ecnt=0;
lcd.print("EAST :"+ String(cnt));
    for(i=cnt;i>0;i--) {
        lcd.setCursor(6,0);
        lcd.print(" ");
        lcd.setCursor(6,0);
        lcd.print(i);

        strip.setPixelColor(0, strip.Color(0, 255, 0));
        strip.setPixelColor(1, strip.Color(255, 0, 0));
    }
}

```

```
strip.setPixelColor(2, strip.Color(255, 0, 0));
strip.setPixelColor(3, strip.Color(255, 0, 0));
strip.show();
scan_rfid();
for(int k=0;k<10;k++)
{
    delay(100);
    scan();
}
}
cnt=5+ncnt;
nt=ncnt;
ncnt=0;
lcd.clear();
lcd.print("NORTH:"+ String(cnt));
for(i=cnt;i>0;i--)
{
    lcd.setCursor(6,0);
    lcd.print(" ");
    lcd.setCursor(6,0);
    lcd.print(i);
    strip.setPixelColor(0, strip.Color(255, 0, 0));
    strip.setPixelColor(1, strip.Color(0, 255, 0));
    strip.setPixelColor(2, strip.Color(255, 0, 0));
    strip.setPixelColor(3, strip.Color(255, 0, 0));
    strip.show();
    scan_rfid();
    scan();
}
```

```

for(int k=0;k<10;k++)
{
    delay(100);
    scan();
}
}

cnt=5+wcnt;

wt=wcnt;

wcnt=0;

lcd.clear();

lcd.print("WEST :"+ String(cnt));

for(i=cnt;i>0;i--)
{
    lcd.setCursor(6,0);

    lcd.print(" ");

    lcd.setCursor(6,0);

    lcd.print(i);

    strip.setPixelColor(0, strip.Color(255, 0, 0));
    strip.setPixelColor(1, strip.Color(255, 0, 0));
    strip.setPixelColor(2, strip.Color(0, 255, 0));
    strip.setPixelColor(3, strip.Color(255, 0, 0));

    strip.show();

    scan_rfid();

    for(int k=0;k<10;k++)
    {
        delay(100);
        scan();
    }
}

```

```

}

cnt=5+scnt;

st=scnt;

scnt=0;

lcd.clear();

lcd.print("SOUTH:"+ String(cnt));


for(i=cnt;i>0;i--)
{
    lcd.setCursor(6,0);
    lcd.print(" ");
    lcd.setCursor(6,0);
    lcd.print(i);
    strip.setPixelColor(0, strip.Color(255, 0, 0));
    strip.setPixelColor(1, strip.Color(255, 0, 0));
    strip.setPixelColor(2, strip.Color(255, 0, 0));
    strip.setPixelColor(3, strip.Color(0, 255, 0));
    strip.show();
    scan_rfid();
    for(int k=0;k<10;k++)
    {
        delay(100);
        scan();
    }
}


mySerial.print("220270,CQPWY5PRWRNE17NH,0,0,src
24G,src@internet,"+String(et) + ","+String(wt)+","+String(nt)+","+String(st)+",0,\n");

```

```

}
void scan_rfid()
{
  if(Serial.available())
  {
    x=Serial.read();
  }

  if (x=='1' || x=='2' || x=='3' || x=='4') //change here the UID of the card/cards that
you want to give access
  {
    if(x=='1')
    {
      digitalWrite(buz,1);
      sts=1;
      delay(1000);
      digitalWrite(buz,0);
      cnt=5+ecnt;
      lcd.clear();
      lcd.print("EAST :"+ String(cnt));
      strip.setPixelColor(0, strip.Color(0, 255, 0));
      strip.setPixelColor(1, strip.Color(255, 0, 0));
      strip.setPixelColor(2, strip.Color(255, 0, 0));
      strip.setPixelColor(3, strip.Color(255, 0, 0));
      strip.show();
    }
    if(x=='2')
    {
      digitalWrite(buz,1);

```

```
sts=1;
delay(1000);
digitalWrite(buz,0);
cnt=5+wcnt;
lcd.clear();
lcd.print("WEST :"+ String(cnt));
    strip.setPixelColor(0, strip.Color(255,  0, 0));
strip.setPixelColor(1, strip.Color(255, 0, 0));
strip.setPixelColor(2, strip.Color(0, 255, 0));
strip.setPixelColor(3, strip.Color(255, 0, 0));
strip.show();
}
if(x=='3')
{
digitalWrite(buz,1);
sts=1;
delay(1000);
digitalWrite(buz,0);
cnt=5+scnt;
lcd.clear();
lcd.print("SOUTH:"+ String(cnt));
    strip.setPixelColor(0, strip.Color(255,  0, 0));
strip.setPixelColor(1, strip.Color(255, 0, 0));
strip.setPixelColor(2, strip.Color(255, 0, 0));
strip.setPixelColor(3, strip.Color(0, 255, 0));
strip.show();
}
if(x=='4')
```



```

{
    digitalWrite(buz,1);

    sts=1;

    delay(1000);

    digitalWrite(buz,0);

    cnt=5+ncnt;

    lcd.clear();

    lcd.print("NORTH:"+ String(cnt));

    strip.setPixelColor(0, strip.Color(255, 0, 0));
    strip.setPixelColor(1, strip.Color(0, 255, 0));
    strip.setPixelColor(2, strip.Color(255, 0, 0));
    strip.setPixelColor(3, strip.Color(255, 0, 0));
    strip.show();
}

x=2;

lcd.setCursor(0,1);

    lcd.print("Emergency Vhcle");
for(i=cnt;i>0;i--)
{
    lcd.setCursor(6,0);

    lcd.print(" ");

    lcd.setCursor(6,0);

    lcd.print(i);

    delay(1000);
}
}
}

```

```
void scan()
{
    if(digitalRead(es)==0)
    {
        ecnt=ecnt+1;
    }
    if(digitalRead(ws)==0)
    {
        wcnt=wcnt+1;
    }
    if(digitalRead(ns)==0)
    {
        ncnt=ncnt+1;
    }
    if(digitalRead(ss)==0)
    {
        scnt=scnt+1;
    }
}
```